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Semiconductor device and electronic apparatus with O-ring reinforcement structure

Abstract

A semiconductor device includes a detector mounted on a base substrate and including a pressure detector to detect pressure, a base portion on the base substrate where the detector is buried, a protruding portion protruding upward from the base portion and including an exposure hole which causes the pressure detector to be exposed upward, and a lid supported by an upper surface of the protruding portion to close the exposure hole. An outer circumferential portion of the lid extends outward from the protruding portion in plan view. The lid includes a slit that is open to an outer side surface and causes the exposure hole to communicate with outside sideward of the semiconductor device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to Japanese Patent Application No. 2020-190317 filed on Nov. 16, 2020 and is a Continuation Application of PCT Application No. PCT/JP2021/033359 filed on Sep. 10, 2021. The entire contents of each application are hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to a waterproof semiconductor device and an electronic apparatus including the semiconductor device.

2. Description of the Related Art

(2) International Publication No. 2019/208127 discloses a waterproof semiconductor device attached to an electronic apparatus such as a pressure measuring device to measure pressure. The semiconductor device includes a base substrate, a detecting element, and a resin package. The detecting element is to detect pressure, and is mounted on the base substrate. The detecting element is electrically connected via a connecting member such as a wire to the base substrate and a circuit element mounted on the base substrate. The resin package is provided to the base substrate. The detecting element and the connecting member are buried in the resin package. The resin package includes an exposure hole for causing a detecting unit of the detecting element to be exposed outside. Via the exposure hole, pressure acts on the detecting unit of the detecting element.

(3) The semiconductor device disclosed in International Publication No. 2019/208127 includes a cylindrical member having a through hole which communicates with the exposure hole. The cylindrical member penetrates through an O ring. With this, the O ring fits in around the cylindrical member. In a state in which the semiconductor device is attached to the electronic apparatus, a gap between a casing of the electronic apparatus and the semiconductor device is sealed by the O ring. With this, the liquid is prevented from entering the inside of the electronic apparatus from outside via the gap.

SUMMARY OF THE INVENTION

(4) When the semiconductor device disclosed in International Publication No. 2019/208127 is attached to the electronic apparatus, the O ring is first fitted in the semiconductor device. Next, the semiconductor device with the O ring fitted therein is attached to the electronic apparatus.

(5) It is desired that fitting the O ring in the semiconductor device can be easily performed. Thus, for example, the O ring is formed so that the inner diameter of the O ring is equal to or larger than the outer diameter of the cylindrical member. This facilitates penetration of the cylindrical member to the O ring. As a result, the O ring can be easily fitted in the cylindrical member.

(6) However, if penetration of the cylindrical member to the O ring is easy as described above, the O ring easily drops off from the cylindrical member. With this, when the semiconductor device with the O ring fitted therein is attached to the electronic apparatus, a possibility that the O ring drops off from the cylindrical member is increased. Thus, there is a fear that attaching the semiconductor device with the O ring fitted therein to the electronic apparatus is difficult.

(7) Therefore, preferred embodiments of the present invention provide semiconductor devices each capable of preventing a fitted O ring from dropping off.

(8) A semiconductor device according to one aspect of a preferred embodiment of the present invention is a semiconductor device which detects pressure, the semiconductor device including a base substrate, a detector mounted on an upper surface of the base substrate and including a pressure detector, a resin package provided on the upper surface of the base substrate, with the detector buried therein, and including an exposure hole which exposes the pressure detector, a lid supported by the resin package so as to close the exposure hole, and a communicating portion which causes the exposure hole to communicate with outside, in which the resin package includes a base portion where the detector is buried, and a protruding portion protruding upward from the base portion and including the exposure hole, in plan view, the base portion extends outward from an outer side surface of the protruding portion, the lid is supported by an upper surface of the protruding portion, the lid includes an extending portion in at least a portion of an outer circumferential portion of the lid, the extending portion extending outward from the outer side surface of the protruding portion in plan view, and the communicating portion is open sideward of the semiconductor device to communicate with outside.

(9) According to preferred embodiments of the present invention, a fitted O ring can be prevented from dropping off.

(10) The above and other elements, features, steps, characteristics and advantages of the present invention will become more apparent from the following detailed description of the preferred embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a semiconductor device according to Preferred Embodiment 1 of the present invention.
- (2) FIG. 2 is a plan view of the semiconductor device according to Preferred Embodiment 1 of the present invention.
- (3) FIG. 3 is a sectional view along an A-A line of FIG. 2.
- (4) FIG. 4 is a sectional view of the semiconductor device of FIG. 3 in a state of being attached to an article of an electronic apparatus.
- (5) FIG. 5 is a plan view of a semiconductor device according to Preferred Embodiment 2 of the present invention.
- (6) FIG. 6 is a plan view of a semiconductor device according to Preferred Embodiment 3 of the present invention.
- (7) FIG. 7 is a plan view of a semiconductor device according to Preferred Embodiment 4 of the present invention.
- (8) FIG. 8 is a plan view of a semiconductor device according to Preferred Embodiment 5 of the present invention.
- (9) FIG. 9 is a sectional view along a B-B line of FIG. 8.
- (10) FIG. 10 is a perspective view of a semiconductor device according to Preferred Embodiment 6 of the present invention.
- (11) FIG. 11 is a sectional view along a C-C line of FIG. 10.
- (12) FIG. 12 is a perspective view of a lid member included in a semiconductor device according to Preferred Embodiment 7 of the present invention from a bottom surface side.
- (13) FIG. 13 is a longitudinal sectional view of the semiconductor device according to Preferred Embodiment 7 of the present invention.
- (14) FIG. 14 is a plan view of a semiconductor device according to Preferred Embodiment 8 of the present invention.
- (15) FIG. 15 is a sectional view along a D-D line of FIG. 14.
- (16) FIG. 16 is a longitudinal sectional view of a modification of the semiconductor device according to Preferred Embodiment 8 of the present invention.
- (17) FIG. 17 is a plan view of a semiconductor device according to Preferred Embodiment 9 of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(18) A semiconductor device according to one aspect of a preferred embodiment of the present invention is a semiconductor device which detects pressure, the semiconductor device including a base substrate, a detector mounted on an upper surface of the base substrate and including a pressure detector, a resin package provided on the upper surface of the base substrate, with the detector buried therein, and including an exposure hole which exposes the pressure detector, a lid supported by the resin package so as to close the exposure hole, and a communicating portion which causes the exposure hole to communicate with outside of the semiconductor device, in which the resin package includes a base portion where the detector is buried, and a protruding portion protruding upward from the base portion and including the exposure hole, in plan view, the base portion extends outward from an outer side surface of the protruding portion, the lid is supported by an upper surface of the protruding portion, the lid includes an extending portion in at

least a portion of an outer circumferential portion of the lid, the extending portion extending outward from the outer side surface of the protruding portion in plan view, and the communicating portion is open sideward of the semiconductor device to communicate with outside.

(19) According to this structure, the lid includes the extending portion. The extending portion extends outward from the outer side surface of the protruding portion. Thus, when the O ring is arranged so that its inner circumferential portion surrounds the protruding portion in plan view, the O ring can be restrained by the extending portion from coming off from above.

(20) If the semiconductor device does not include the lid, there is a fear that a foreign matter falling from above may make contact with the pressure detector via the exposure hole to influence detection of pressure by the pressure detector. With this structure, since the exposure hole is covered with the lid, a foreign matter falling from above is prevented from entering the exposure hole.

(21) To cause pressure from outside to act on the pressure detector while the exposure hole is closed with the lid, it is thought that a through hole penetrating through the lid in an up-down direction to cause the outside and the exposure hole to communicate with each other is provided. However, when a foreign matter falling from above closes the through hole, it is not possible to cause pressure from outside to act on the pressure detector. However, according to this structure, pressure from outside is caused to act on the pressure detecting from a side of the semiconductor device via the communicating portion and the exposure hole. Thus, a possibility that a foreign matter falling from above closes the communicating portion can be decreased.

(22) The communicating portion may have a space defined by a slit located inward of the outer circumferential portion of the lid in plan view.

(23) According to this structure, by providing only the slit in the lid, the function of the communicating portion can be achieved.

(24) According to this structure, the lid is open by the slit not only sideward but also upward. Thus, the communicating portion can cause the exposure hole and the outside of the semiconductor device to communicate with each other not only sideward but also upward of the semiconductor device. Since this facilitates circulation of the fluid from outside to the exposure hole, responsiveness of the pressure detector which detects pressure of the fluid can be improved. Also, even if a portion above the slit is closed with a foreign matter, communication from a side of the semiconductor device to the exposure hole can be maintained.

(25) The slit may be at a position outside of the pressure detector in plan view.

(26) According to this structure, a possibility that a foreign matter falling from above makes contact with the pressure detector via the slit can be decreased.

(27) The slit may include a wide portion at least a portion of which overlaps the exposure hole in plan view, and a narrow portion connected to the wide portion and not overlapping the exposure hole in plan view.

(28) According to this structure, circulation of the fluid between above the lid and the exposure hole is facilitated by the wide portion. Thus, responsiveness of the pressure detector to detect pressure of the fluid can be improved.

(29) According to this structure, the width of the slit is narrow at a portion not overlapping the exposure hole in plan view, that is, a portion where the lid and the upper surface of the protruding portion are in contact with each other. Thus, the contact area between the lid and the protruding portion can be increased. This stabilizes the lid.

(30) The lid may include three or more of the slits.

(31) According to this structure, when the lid is supported by the upper surface of the protruding portion, the outer side surface of the protruding portion is visually recognized at three or more locations via the respective three or more slits provided. With this, positioning of the lid with respect to the protruding portion can be accurately performed.

(32) A semiconductor device according to one aspect of a preferred embodiment of the present

invention may include an adhering portion interposed partially between the upper surface of the protruding portion and the lid to fix the lid to the protruding portion. The communicating portion may include a space between the upper surface of the protruding portion and the lid which are opposed to each other in a portion between the upper surface of the protruding portion and the lid where the adhering portion is not interposed.

(33) According to this structure, when the lid is adhered to the protruding portion, by only providing an area where no adhesive is applied between the lid and the protruding portion, that area can be made function as the communicating portion.

(34) The protruding portion may include a first concave portion recessed downward from the upper surface of the protruding portion. The communicating portion may include a space defined by the first concave portion.

(35) According to this structure, only by providing the first concave portion in the protruding portion, the function of the communicating portion can be achieved.

(36) The lid may include a second concave portion recessed upward from a lower surface of the lid. The communicating portion may include a space defined by the second concave portion.

(37) According to this structure, only by providing the second concave portion in the lid, the function of the communicating portion can be achieved.

(38) A semiconductor device according to one aspect of a preferred embodiment of the present invention may include at least two of a slit located inward from the outer circumferential portion of the lid in plan view, an adhering portion interposed partially between the upper surface of the protruding portion and the lid to fix the lid to the protruding portion, a first concave portion recessed downward from the upper surface of the protruding portion and a second concave portion recessed upward from a lower surface of the lid, in which when the semiconductor device includes the slit, the communicating portion may include a first space defined by the slit, when the semiconductor device includes the adhering portion, the communicating portion may include a second space between the upper surface of the protruding portion and the lid which are opposed to each other in a portion between the upper surface of the protruding portion and the lid where the adhering portion is not provided, when the semiconductor device includes the first concave portion, the communicating portion may include a third space defined by the first concave portion, when the semiconductor device includes the second concave portion, the communicating portion may include a fourth space defined by the second concave portion, and in plan view, the first space, the second space, the third space, and the fourth space may overlap one another.

(39) According to this structure, the cross-sectional area of the communicating portion can be increased. Since this facilitates circulation of the fluid in the communicating portion, responsiveness of the pressure detector to detect pressure of the fluid can be improved.

(40) A semiconductor device according to one aspect of a preferred embodiment of the present invention may include at least two of a slit located inward from the outer circumferential portion of the lid in plan view, an adhering portion interposed partially between the upper surface of the protruding portion and the lid to fix the lid to the protruding portion, a first concave portion recessed downward from the upper surface of the protruding portion, and a second concave portion recessed upward from a lower surface of the lid, in which when the semiconductor device includes the slit, the communicating portion may include a first space defined by the slit, when the semiconductor device includes the adhering portion, the communicating portion may include a second space between the upper surface of the protruding portion and the lid which are opposed to each other in a portion between the upper surface of the protruding portion and the lid where the adhering portion is not provided, when the semiconductor device includes the first concave portion, the communicating portion may include a third space defined by the first concave portion, when the semiconductor device includes the second concave portion, the communicating portion may include a fourth space defined by the second concave portion, and in plan view, the first space, the second space, the third space, and the fourth space do not have to overlap one another.

(41) According to this structure, the number of paths through which the fluid circulates can be increased. Since this facilitates circulation of the fluid in the communicating portion, responsiveness of the pressure detector to detect pressure of the fluid can be improved.

(42) A semiconductor device according to one aspect of a preferred embodiment of the present invention may include a circuit element mounted on the upper surface of the base substrate and a connector to electrically connect the circuit element and the detector together.

(43) An electronic apparatus according to one aspect of a preferred embodiment of the present invention may include the above-described semiconductor, an O ring surrounding the protruding portion of the resin package of the semiconductor device in plan view and including an inner circumferential portion in contact with the outer side surface of the protruding portion, and a casing to which the semiconductor device is attached.

(44) According to this structure, in the electronic apparatus including the semiconductor device, the O ring can be restrained by the extending portion from coming off from above.

Preferred Embodiment 1

(45) FIG. 1 is a perspective view of a semiconductor device **10** according to Preferred Embodiment 1 of the present invention. FIG. 2 is a plan view of the semiconductor device **10** according to Preferred Embodiment 1 of the present invention. FIG. 3 is a sectional view along an A-A line of FIG. 2. FIG. 4 is a sectional view of the semiconductor device **10** of FIG. 3 in a state of being attached to an article of an electronic apparatus **1**. An X-Y-Z orthogonal coordinate system depicted in these drawings and the drawings described below are for ease of understanding of preferred embodiments of the present invention and do not limit the present invention.

(46) A semiconductor device **10** is a pressure sensor which detects pressure. As depicted in FIG. 1 to FIG. 3, the semiconductor device **10** includes a base substrate **12**, a circuit element **14** mounted on the base substrate **12**, a detecting element **16** mounted on the base substrate **12**, a resin package **22**, and a lid member **26**.

(47) In Preferred Embodiment 1, the base substrate **12** is a rigid substrate such as a glass epoxy substrate or ceramic substrate, but is not limited to this. For example, the base substrate **12** may be a lead frame.

(48) As depicted in FIG. 3, the circuit element **14** and the detecting element **16** are mounted on an upper surface **12a** of the base substrate **12**. The base substrate **12** includes a pad **12b** on the upper surface **12a**. While one pad **12b** is depicted in FIG. 3, the number of pads **12b** is not limited to one. The pad **12b** is electrically connected via a bonding wire **18** to the circuit element **14**. The base substrate **12** includes an external connection terminal **12c** on a lower surface **12d**, which is a back surface of the upper surface **12a**. The semiconductor device **10** is electrically connected to another external device (not depicted) via the external connection terminal **12c**.

(49) The circuit element **14** includes an upper surface **14a** and a lower surface **14b**, which is a back surface of the upper surface **14a**. In Preferred Embodiment 1, the circuit element **14** is an element including an application specific integrated circuit (ASIC). In Preferred Embodiment 1, the lower surface **14b** of the circuit element **14** is adhered to the upper surface **12a** of the base substrate **12** with a die attach film, a die attach material, or the like.

(50) The circuit element **14** includes a first pad **14c** on the upper surface **14a**. While only one first pad **14c** is depicted in FIG. 3, the number of first pads **14c** is not limited to one. The first pad **14c** is electrically connected via the bonding wire **18** to the pad **12b** of the base substrate **12**.

(51) The circuit element **14** includes a second pad **14d** on the upper surface **14a**, separately from the first pad **14c**. While one second pad **14d** is depicted in FIG. 3, the number of second pads **14d** is not limited to one. The second pad **14d** is electrically connected via a bonding wire **20** to the detecting element **16**. The bonding wire **20** is one example of a connecting member.

(52) The circuit element **14** includes a signal processing circuit which processes a signal outputted from the detecting element **16** and outputs the processed signal to the base substrate **12**. For example, in Preferred Embodiment 1, the circuit element **14** includes a converter, a filter, a

temperature sensor, a processor, a memory, and so forth. The converter converts a voltage signal outputted from the detecting element **16** to a digital signal. The filter filters the digital signal from the converter. The temperature sensor detects temperature. The processor corrects the filtered digital signal based on the detected temperature of the temperature sensor. The memory stores a correction coefficient for use in correcting the digital signal using the detected temperature, and so forth.

(53) In Preferred Embodiment 1, the detecting element **16** is a pressure sensor element for measuring pressure. The detecting element **16** includes an upper surface **16a** and a lower surface **16b**, which is a back surface of the upper surface **16a**. The detecting element **16** is, for example, a piezoresistance-type pressure sensor element or an electrostatic capacitance-type pressure sensor element, and a micro electro mechanical systems (MEMS) element. In Preferred Embodiment 1, the lower surface **16b** of the detecting element **16** is adhered to the upper surface **12a** of the base substrate **12** with a die attach film, a die attach material, or the like. Note that while the detecting element **16** is mounted on the upper surface **12a** of the base substrate **12** in Preferred Embodiment 1, the detecting element **16** may be mounted on the circuit element **14**.

(54) The detecting element **16** includes a pad **16c** on the upper surface **16a**. While one pad **16c** is depicted in FIG. 3, the number of pads **16c** is not limited to one. The pad **16c** is electrically connected via the bonding wire **20** to the second pad **14d** of the circuit element **14**. That is, the bonding wire **20** of the detecting element **16** is electrically connected via the circuit element **14** and the bonding wire **18** to the base substrate **12**.

(55) The detecting element **16** includes a detecting unit **16d**, on which pressure acts, on the upper surface **16a**. That is, the detecting unit **16d** detects pressure. In Preferred Embodiment 1, the detecting unit **16d** of the detecting element **16** as a pressure sensor element is a membrane or diaphragm which receives pressure. Note that the detecting unit **16d** includes, for example, a passivation film to be made waterproof.

(56) The resin package **22** is a package fabricated by molding hard resin such as, for example, thermosetting resin, on the upper surface **12a** of the base substrate **12**. That is, as depicted in FIG. 3, the resin package **22** is provided on the upper surface **12a** of the base substrate **12**. A portion (in particular, the pad **12b**) of the upper surface **12a** of the base substrate **12** is protected and made waterproof by being covered with the resin package **22**.

(57) As depicted in FIG. 3, the circuit element **14**, the detecting element **16**, and the bonding wires **18** and **20** are buried in the resin package **22**. Buried inside the resin package **22**, the circuit element **14** (in particular, the first pad **14c** and the second pad **14d**), the detecting element **16** (in particular, the pad **16c**), and the bonding wires **18** and **20** are made waterproof so as to be protected.

(58) As described above, electrical connection between the base substrate **12** and the circuit element **14** and electrical connection between the circuit element **14** and the detecting element **16** are made waterproof by the resin package **22**.

(59) As depicted in FIG. 1 and FIG. 3, the resin package **22** includes a base portion **23** and a protruding portion **24**.

(60) The base portion **23** configures a base substrate **12** side (Z-axis negative direction side) of the resin package **22**. As depicted in FIG. 3, a lower surface **23a** of the base portion **23** is in contact with the upper surface **12a** of the base substrate **12**.

(61) The circuit element **14**, the detecting element **16**, and the bonding wires **18** and **20** are buried in the base portion **23**.

(62) The protruding portion **24** configures a side (Z-axis positive direction side), of the resin package **22**, opposite to the base substrate **12** with respect to the base portion **23**. The protruding portion **24** protrudes from the base portion **23** so as to be away from the base substrate **12**. In other words, the protruding portion **24** protrudes from the base portion **23** to the Z-axis position direction (upward).

(63) As depicted in FIG. 1 and FIG. 2, the protruding portion **24** is provided at a center portion of

the base portion **23** in plan view. In other words, in plan view, the base portion **23** extends outward from an outer side surface **24a** of the protruding portion **24**.

(64) The protruding portion **24** includes an exposure hole **22a**. The exposure hole **22a** penetrates through the protruding portion **24** to the Z-axis direction (up-down direction). The protruding portion **24** is cylindrical.

(65) As depicted in FIG. 3, a portion of the upper surface **16a** of the detecting element **16** and the detecting unit **16d** face a lower end portion of the exposure hole **22a**. That is, a portion of the upper surface **16a** of the detecting element **16** and the detecting unit **16d** are exposed via the exposure hole **22a** outside the resin package **22**. That is, the exposure hole **22a** exposes the detecting unit **16d** of the detecting element **16** outside the resin package **22**. An opening of the exposure hole **22a** is provided on an upper surface **24b** (surface in the Z-axis position direction) of the protruding portion **24**. That is, the resin package **22** exposes the detecting unit **16d** of the detecting element **16** upward. Via a slit **27** of the lid member **26** described below and the exposure hole **22a**, pressure acts on the detecting unit **16d** of the detecting element **16** from outside the semiconductor device **10**, and the detecting element **16** can measure that pressure.

(66) As depicted in FIG. 2, in plan view, the protruding portion **24** is positioned around the detecting unit **16d** of the detecting element **16**. That is, in plan view, the protruding portion **24** is positioned outside the detecting unit **16d**.

(67) Note that in the detecting element **16**, the detecting unit **16d**, which is a portion exposed outside by the exposure hole **22a**, is made waterproof by, for example, a passivation film. Also, in Preferred Embodiment 1, the shape of the opening of the exposure hole **22a** is a circular or substantially circular shape when viewed from a top surface (Z-axis direction) as depicted in FIG. 1 and FIG. 2 and, as depicted in FIG. 3, is a tapered or substantially tapered shape with its cross-sectional area along an XY plane decreasing from the opening as being closer to the detecting unit **16d** of the detecting element **16**, but may be another shape. The exposure hole **22a** may be any type or shape as long as it functions as a pressure introduction hole which introduces pressure to the detecting unit **16d**.

(68) As depicted in FIG. 1 to FIG. 3, the lid member **26** is supported by the upper surface **24b** of the protruding portion **24** of the resin package **22** so as to close the exposure hole **22a**. In Preferred Embodiment 1, the lid member **26** is attached to the upper surface **24b** of the protruding portion **24** with an adhesive (not depicted). In Preferred Embodiment 1, the adhesive is applied over the entire circumference of the upper surface **24b** of the protruding portion **24** in plan view. Note that the adhesive is depicted only in FIG. 8 and FIG. 9 of Preferred Embodiment 5 described below and its depiction is omitted in each of the other drawings.

(69) As depicted in FIG. 1 and FIG. 2, in Preferred Embodiment 1, the lid member **26** has a disk shape or substantially disk shape. Note that the shape of the lid member **26** is not limited to a disk shape and may be, for example, a rectangular or substantially rectangular parallelepiped shape.

(70) In Preferred Embodiment 1, the lid member **26** is configured of resin such as liquid-crystal polymer. Note that the lid member **26** may be configured of a material not limited to resin but may be configured of metal such as, for example, stainless steel (SUS). In Preferred Embodiment 1, the thickness (length in the Z-axis direction) of the lid member **26** is about 50 to about 200 (μm), and the diameter of the lid member **26** is about 2 to about 4 (mm), for example.

(71) The lid member **26** includes a slit **27**. In plan view, the slit **27** extends from an outer circumferential portion **26a** of the lid member **26** toward a center portion **26b** of the lid member **26**. In other words, in plan view, the slit **27** extends from the outer circumferential portion **26a** of the lid member **26** toward inside the lid member **26**. The outer circumferential portion **26a** of the lid member **26** is a portion formed of an outer side surface **26c** of the lid member **26** and a nearby portion of the outer side surface **26c**. The center portion **26b** of the lid member **26** is a portion of the lid member **26** overlapping the exposure hole **22a** in plan view.

(72) In Preferred Embodiment 1, as depicted in FIG. 1 to FIG. 3, the slit **27** extends from the outer

side surface **26c** of the lid member **26** to the center portion **26b** of the lid member **26**. That is, in plan view, a portion (depth portion) of the slit **27** overlaps the exposure hole **22a**. However, in Preferred Embodiment 1, as depicted in FIG. 2, in plan view, the depth portion of the slit **27** does not overlap the detecting unit **16d** of the detecting element **16**. That is, in Preferred Embodiment 1, in plan view, the slit **27** is at a position out of the detecting unit **16d** of the detecting element **16**. Note that the slit **27** may overlap the detecting unit **16d** of the detecting element **16** in plan view.

(73) The slit **27** opens upward (Z-axis positive direction side), downward (Z-axis negative direction side), and outward (radial direction radially extending from the center toward the outer circumferential portion **26a** of the lid member **26** in plan view) of the lid member **26**. The slit **27** opens downward, thereby communicating with the exposure hole **22a**. The slit **27** opens upward and outward of the lid member **26**, thereby communicating with the outside of the semiconductor device **10**. With this, the slit **27** causes the outside of the semiconductor device **10** and the exposure hole **22a** to communicate with each other.

(74) In Preferred Embodiment 1, a width W (refer to FIG. 2) of the slit **27** is, for example, about 50 to about 500 (μm), but is not limited this.

(75) The slit **27** (in detail, a space defined by the slit **27**) is one example of a communicating portion. The space including the slit **27** is one example of a first space. It can also be said that inner side surfaces **27a**, **27b**, and **27c**, of the lid member **26**, configuring the slit **27** are one example of the communicating portion.

(76) In Preferred Embodiment 1, as depicted in FIG. 2, in plan view, the outer circumferential portion **26a** of the lid member **26** extends from the outer side surface **24a** of the protruding portion **24** outward in the radial direction. In Preferred Embodiment 1, the outer circumferential portion **26a** of the lid member **26** corresponds to an extending portion. That is, in Preferred Embodiment 1, the entirety of the outer circumferential portion **26a** of the lid member **26** is an extending portion extending from the outer side surface **24a** of the protruding portion **24** outward in plan view.

(77) FIG. 4 is a sectional view of the semiconductor device of FIG. 3 in a state of being attached to an article of an electronic apparatus. As depicted in FIG. 3 and FIG. 4, the resin package **22** is configured so as to support an O ring OR.

(78) As depicted in FIG. 4, the semiconductor device **10** is used as attached to a casing **2** of an electronic apparatus **1**. The electronic apparatus **1** is, for example, a pressure measuring device, and includes the semiconductor device **10**, the O ring OR, and the casing **2**.

(79) The O ring OR has an annular or substantially annular shape. The inner diameter of the O ring OR is configured to be equal to or substantially equal to the outer diameter of the protruding portion **24** of the resin package **22**. The O ring OR includes a material or structure that is easy to compress and deform, such as nitrile rubber.

(80) As indicated by broken lines in FIG. 3, the O ring OR is inserted from above to the protruding portion **24** of the resin package **22** before the lid member **26** is attached. The O ring OR inserted to the protruding portion **24** is arranged so as to surround the protruding portion **24** in plan view. In Preferred Embodiment 1, the inner diameter of the O ring OR is configured to be larger than the outer diameter of the protruding portion **24**. With this, the above-described insertion is easily performed. After insertion of the O ring OR to the protruding portion **24**, the lid member **26** is adhered to the upper surface **24b** of the protruding portion **24**.

(81) As depicted in FIG. 3, the O ring OR inserted to the protruding portion **24** is restrained by the outer circumferential portion **26a** of the lid member **26** from coming off from an upper end portion of the protruding portion **24**. The O ring OR inserted to the protruding portion **24** is restricted by the base portion **23** of the resin package **22** from coming off from a lower end portion of the protruding portion **24**.

(82) As depicted in FIG. 4, the casing **2** includes an inner space **2a** where the semiconductor device **10** with the O ring OR inserted thereto is arranged. The semiconductor device **10** arranged in the inner space **2a** is attached to the casing **2** by known methods or materials (not depicted) such as

fitting.

(83) The casing **2** has a through hole **2b**. The through hole **2b** causes the outside and the inner space **2a** of the casing **2** to communicate with each other. The exposure hole **22a** of the semiconductor device **10** communicates with the outside of the electronic apparatus **1** via the slit **27** of the lid member **26**, the inner space **2a** of the casing **2**, and the through hole **2b** of the casing **2**. With this, external pressure of the electronic apparatus **1** can be measured by the semiconductor device **10**.

(84) To seal a gap between the semiconductor device **10** arranged as described above and an inner circumferential surface **2c** configuring the inner space **2a** of the casing **2**, the O ring OR is arranged in this gap.

(85) In the state in which the semiconductor device **10** is arranged in the inner space **2a** of the casing **2**, a gap G between the outer side surface **24a** of the protruding portion **24** and the inner circumferential surface **2c** configuring the inner space **2a** is shorter than a diameter R (refer to FIG. 3) of the O ring OR. Thus, when the semiconductor device **10** is arranged in the inner space **2a**, the O ring OR is compressed and deformed (refer to FIG. 4). With this, an inner circumferential portion ORa of the O ring OR makes contact with the outer side surface **24a** of the protruding portion **24**, and an outer circumferential portion ORb of the O ring OR makes contact with the inner circumferential surface **2c** configuring the inner space **2a**. As a result, the gap between the semiconductor device **10** and the casing **2** is sealed, and liquid is prevented by the O ring OR from entering the inside of the electronic apparatus **1** from the outside of the electronic apparatus **1** via the gap.

(86) According to Preferred Embodiment 1, the outer circumferential portion **26a** of the lid member **26** is an extending portion extended from the outer side surface **24a** of the protruding portion **24** outward in the radial direction. Thus, when the O ring OR is arranged so as to surround the protruding portion **24** in plan view, the O ring OR is restrained by the extending portion from coming off from above.

(87) If the semiconductor device **10** does not include the lid member **26**, there is a fear that a foreign matter falling from above may make contact with the detecting unit **16d** via the exposure hole **22a** to influence detection of pressure by the detecting unit **16d**. According to Preferred Embodiment 1, since the exposure hole **22a** is covered with the lid member **26**, a foreign matter falling from above is prevented from entering the exposure hole **22a**.

(88) To cause pressure from outside to act on the detecting unit **16d** while the exposure hole **22a** is closed with the lid member **26**, it is thought that a through hole penetrating through the lid member **26** in the up-down direction (Z-axis direction) to cause the outside and the exposure hole **22a** to communicate with each other is provided. However, when the through hole is closed with a foreign matter falling from above, it is not possible to cause pressure from outside to act on the detecting unit **16d**. However, according to Preferred Embodiment 1, pressure from outside is caused to act on the detecting unit **16d** from sideward of the semiconductor device **10** via the slit **27** and the exposure hole **22a**. Thus, a possibility that a foreign matter falling from above reach the exposure hole **22a** can be decreased.

(89) According to Preferred Embodiment 1, only with the slit **27** provided in the lid member **26**, the function of the communicating portion can be achieved.

(90) According to Preferred Embodiment 1, the lid member **26** is open by the slit **27** not only sideward but also upward. Thus, the slit **27** can cause the exposure hole **22a** and the outside of the semiconductor device **10** to communicate with each other not only sideward but also upward of the semiconductor device **10**. Since this facilitates circulation of the fluid from outside to the exposure hole **22a**, responsiveness of the detecting unit **16d** which detects pressure of the fluid can be improved. Also, even if a portion above the slit **27** is closed with a foreign matter, communication from sideward of the semiconductor device **10** to the exposure hole **22a** can be maintained.

(91) According to Preferred Embodiment 1, a possibility that a foreign matter falling from above makes contact with the detecting unit **16d** via the slit **27** can be decreased.

(92) According to Preferred Embodiment 1, in the electronic apparatus **1** including the semiconductor device **10**, the O ring OR can be restrained by the outer circumferential portion **26a** of the lid member **26** from coming off from above.

(93) The structure of the shape of the resin package **22** and so forth is not limited to the above-described structure (structure depicted in FIG. **1** to FIG. **4**). For example, while the base portion **23** of the resin package **22** has a rectangular or substantially rectangular shape in plan view in Preferred Embodiment 1, the base portion **23** may have a shape other than a rectangle (for example, circular shape) in plan view.

(94) In Preferred Embodiment 1, the circuit element **14** is electrically connected via the bonding wire **18** to the base substrate **12**, and the detecting element **16** is electrically connected via the bonding wire **20** to the circuit element **14**. However, the connection mode among the base substrate **12**, the circuit element **14**, and the detecting element **16** is not limited to this. For example, as with the circuit element **14**, the detecting element **16** may be connected via a bonding wire to the base substrate **12**. In this case, the detecting element **16** is electrically connected via the base substrate **12** to the circuit element **14**.

(95) In Preferred Embodiment 1, the base substrate **12**, the circuit element **14**, and the detecting element **16** were each electrically connected via a bonding wire. However, the base substrate **12**, the circuit element **14**, and the detecting element **16** may be electrically connected by methods or structures other than a bonding wire. For example, the circuit element **14** may be mounted on the base substrate **12** with a flip chip. In this case, the circuit element **14** and the base substrate **12** are electrically connected together by soldering.

Preferred Embodiment 2

(96) FIG. **5** is a plan view of a semiconductor device according to Preferred Embodiment 2 of the present invention. A semiconductor device **10A** according to Preferred Embodiment 2 is different from the semiconductor device **10** according to Preferred Embodiment 1 in that the lid member **26** includes a slit **28** in place of the slit **27**.

(97) As depicted in FIG. **5**, as with the slit **27** depicted in FIG. **2**, the slit **28** is formed from the outer circumferential portion **26a** of the lid member **26** toward the center portion **26b** of the lid member **26**.

(98) The slit **28** includes a wide portion **28A** and a narrow portion **28B**. The wide portion **28A** is longer than the narrow portion **28B** in a width direction. In Preferred Embodiment 2, the width direction is the X-axis direction.

(99) The wide portion **28A** configures a depth portion of the slit **28**. At least a portion of the wide portion **28A** overlaps the exposure hole **22a** in plan view. In Preferred Embodiment 2, the wide portion **28A** does not overlap the detecting unit **16d** of the detecting element **16** in plan view, but may overlap it.

(100) The narrow portion **28B** configures a portion other than the depth portion of the slit **28**. One end portion of the narrow portion **28B** is connected to the wide portion **28A**. The other end portion of the narrow portion **28B** opens to the outer side surface **26c** of the lid member **26**. The narrow portion **28B** is positioned outside the exposure hole **22a** in a radial direction of the lid member **26** in plan view. That is, the narrow portion **28B** does not overlap the exposure hole **22a** in plan view.

(101) The slit **28** (in detail, a space defined by the slit **28**) is one example of the communicating portion. The space defined by the slit **28** is one example of the first space. It can also be said that an inner side surface **28a**, of the lid member **26**, configuring the slit **28** is one example of the communicating portion.

(102) In Preferred Embodiment 2, the slit **28** has a tapered shape which becomes narrower toward outside in the radial direction of the lid member **26**. However, the shape of the slit **28** is not limited to the tapered shape. For example, as indicated by one-dot-chain lines in FIG. **5**, the slit **28** may include a step at a boundary portion between the wide portion **28A** and the narrow portion **28B**.

(103) According to Preferred Embodiment 2, circulation of the fluid between above the lid member

26 and the exposure hole **22a** is facilitated by the wide portion **28A**. Thus, responsiveness of the detecting unit **16d** which detects pressure of the fluid can be improved.

(104) According to Preferred Embodiment 2, the width of the slit **28** is narrow at a portion not overlapping the exposure hole **22a** in plan view, that is, a portion where the lid member **26** and the upper surface **24b** of the protruding portion **24** are in contact with each other. Thus, the contact area between the lid member **26** and the protruding portion **24** can be increased. This stabilizes the lid member **26**.

Preferred Embodiment 3

(105) FIG. **6** is a plan view of a semiconductor device according to Preferred Embodiment 3 of the present invention. A semiconductor device **10B** according to Preferred Embodiment 3 is different from the semiconductor device **10** according to Preferred Embodiment 1 in that the lid member **26** includes a plurality of slits **27**.

(106) As depicted in FIG. **6**, the lid member **26** of the semiconductor device **10B** includes four slits **27**. The four slits **27** are equidistantly spaced in a circumferential direction CD of the lid member **26**. In other words, the four slits **27** are arranged with 90-degree pitches in the circumferential direction CD.

(107) Note that the number of slits **27** included in the lid member **26** is not limited to four. Also, the plurality of respective slits **27** may be arranged with different pitches in the circumferential direction CD of the lid member **26**. Also, the width of each slit **27** may be equal as depicted in FIG. **6** or may be different. Also, the width of each slit **27** is not limited to the width depicted in FIG. **6**.

(108) According to Preferred Embodiment 3, when the semiconductor device **10B** includes three or more slits **27** and the lid member **26** is supported by the upper surface **24b** of the protruding portion **24**, the outer side surface **24a** of the protruding portion **24** is visually recognized at three or more locations via the respective three or more slits **27** provided. With this, positioning of the lid member **26** with respect to the protruding portion **24** can be accurately performed.

Preferred Embodiment 4

(109) FIG. **7** is a plan view of a semiconductor device according to Preferred Embodiment 4 of the present invention. A semiconductor device **10C** according to Preferred Embodiment 4 is different from the semiconductor device **10** according to Preferred Embodiment 1 in that the semiconductor device **10C** includes a lid member **29** in place of the lid member **26**.

(110) As with the lid member **26** (refer to FIG. **2**) of the semiconductor device **10** of Preferred Embodiment 1, the lid member **29** is supported by the upper surface **24b** of the protruding portion **24** of the resin package **22** so as to close the exposure hole **22a**, and has the slit **27**. The slit **27** has the same structure as that of the lid member **26** (refer to FIG. **2**) of the semiconductor device **10** of Preferred Embodiment 1.

(111) In Preferred Embodiment 3, the lid member **29** has a disk shape or substantially disk shape. The diameter of the lid member **29** is equal to or smaller than the outer diameter of the protruding portion **24**. In FIG. **7**, the diameter of the lid member **29** is equal to the outer diameter of the protruding portion **24**. The lid member **29** includes four extending portions **29A**. Each extending portion **29A** extends from an outer side surface **29c** of the lid member **29** toward outside in a radial direction of the lid member **29**. The radial direction of the lid member **29** is a direction radially extending from the center of the lid member **29** toward an outer circumferential portion **29a** in plan view.

(112) The four extending portions **29A** are equidistantly spaced in the circumferential direction CD of the lid member **29**. In other words, the four extending portions **29A** are arranged with 90-degree pitches in the circumferential direction CD. In Preferred Embodiment 4, a nearby portion of an outer side surface **29c** in the outer circumferential portion **29a** of the lid member **29** includes not only an inner portion in the radial direction with respect to the outer side surface **29c** in plan view but also the four extending portions **29A**, which are outer portions in the radial direction with respect to the outer side surface **29c** in plan view. That is, in Preferred Embodiment 4, the lid

member **29** includes the extending portions **29A** at a portion of the outer circumferential portion **29a**.

(113) Note that the number of extending portions **29A** included in the lid member **29** is not limited to four. Also, the plurality of respective extending portions **29A** may be arranged with different pitches in the circumferential direction CD of the lid member **29**. Also, the length of each extending portion **29A** in the circumferential direction CD is not limited to the length depicted in FIG. 7.

Preferred Embodiment 5

(114) FIG. **8** is a plan view of a semiconductor device according to Preferred Embodiment 5 of the present invention. FIG. **9** is a sectional view along a B-B line of FIG. **8**. A semiconductor device **10D** according to Preferred Embodiment 5 is different from the semiconductor device **10** according to Preferred Embodiment 1 in that the lid member **26** does not include the slit **27** and there are portions between the lid member **26** and the upper surface **24b** of the protruding portion **24** where an adhesive is not interposed.

(115) As depicted in FIG. **8**, the lid member **26** of the semiconductor device **10D** does not include the slit **27** as depicted in FIG. **2**.

(116) As depicted in FIG. **9**, the lid member **26** is attached to the upper surface **24b** of the protruding portion **24** with an adhesive. An adhering portion **30** configured of the adhesive is interposed between the protruding portion **24** and the lid member **26**. In Preferred Embodiment 5, at two locations of the upper surface **24b** of the protruding portion **24** in the circumferential direction CD in plan view, there are portions to which the adhesive is not applied. In other words, in Preferred Embodiment 5, between a lower surface **26d** of the lid member **26** and the upper surface **24b** of the protruding portion **24**, there are portions where the adhering portion **30** is not interposed. In those portions, spaces **31** are provided between the lower surface **26d** of the lid member **26** and the upper surface **24b** of the protruding portion **24**. That is, the lower surface **26d** of the lid member **26** and the upper surface **24b** of the protruding portion **24** are opposed to each other via the spaces **31** in the Z-axis direction.

(117) As depicted in FIG. **8**, each space **31** extends from the outer side surface **24a** to an inner side surface **24c** of the protruding portion **24**. The inner side surface **24c** of the protruding portion **24** is a surface configuring the exposure hole **22a**. That is, via each space **31**, the exposure hole **22a** and the outside of the semiconductor device **10D** communicate with each other.

(118) In Preferred Embodiment 5, in plan view, the two spaces **31** are opposed to each other across the detecting unit **16d** of the detecting element **16**. In other words, the two spaces **31** are arranged with 180-degree pitches in the circumferential direction CD.

(119) Note that the number of spaces **31** between the lower surface **26d** of the lid member **26** and the upper surface **24b** of the protruding portion **24** is not limited to two. Also, the position where each space **31** is located is not limited to the position depicted in FIG. **8**. Also, the two spaces **31** may be arranged in the circumferential direction CD with pitches other than 180-degree pitches. For example, between the two spaces **31**, a narrower pitch in the circumferential direction CD may be 90 degrees and a wider pitch in the circumferential direction CD may be 270 degrees. Also, when three or more spaces **31** are provided, the respective spaces **31** may be equidistantly arranged in the circumferential direction CD or may be arranged with different pitches. Also, the length of each space **31** in the circumferential direction CD is not limited to the length depicted in FIG. **8**. Also, the length of each space **31** in the circumferential direction CD may be equal as depicted in FIG. **8** or different.

(120) The space **31** is one example of the communicating portion. Also, the space **31** is one example of a second space. It can also be said that the lower surface **26d** of the lid member **26**, the upper surface **24b** of the protruding portion **24**, and side end portions **30a** of the adhering portion **30**, which configure the space **31**, are one example of the communicating portion.

(121) According to Preferred Embodiment 5, when the lid member **26** is adhered to the protruding portion **24**, by only providing the space **31** where no adhesive is applied partially between the lid

member **26** and the protruding portion **24**, the space **31** can be made function as the communicating portion.

Preferred Embodiment 6

(122) FIG. **10** is a perspective view of a semiconductor device according to Preferred Embodiment 6 of the present invention. FIG. **11** is a sectional view along a C-C line of FIG. **10**. Note that depiction of the lid member **26** is omitted in FIG. **10**. A semiconductor device **10E** according to Preferred Embodiment 6 is different from the semiconductor device **10** according to Preferred Embodiment 1 in that the lid member **26** does not include the slit **27** and the protruding portion **24** of the resin package **22** includes concave portions **24A**.

(123) As depicted in FIG. **11**, the lid member **26** of the semiconductor device **10E** does not include the slit **27** as depicted in FIG. **2**.

(124) As depicted in FIG. **10**, the protruding portion **24** of the resin package **22** includes two concave portions **24A**. The concave portions **24A** each are one example of a first concave portion. Each concave portion **24A** is formed with the upper surface **24b** of the protruding portion **24** recessed downward. Each concave portion **24A** is formed from the outer side surface **24a** to the inner side surface **24c** of the protruding portion **24**. That is, as depicted in FIG. **10** and FIG. **11**, via each concave portion **24A**, the exposure hole **22a** and the outside of the semiconductor device **10E** communicate with each other.

(125) In Preferred Embodiment 6, as depicted in FIG. **10**, the two concave portions **24A** are positioned on one straight line in the Y-axis direction. In other words, the two concave portions **24A** are arranged with 180-degree pitches in the circumferential direction CD.

(126) Note that the number of concave portions **24A** included in the protruding portion **24** is not limited to two. Also, the position where each concave portion **24A** is formed is not limited to the position depicted in FIG. **10**. Also, the two concave portions **24A** may be arranged in the circumferential direction CD with pitches other than 180-degree pitches. For example, between the two concave portions **24A**, a narrower pitch in the circumferential direction CD may be 90 degrees and a wider pitch in the circumferential direction CD may be 270 degrees. Also, when three or more concave portions **24A** are provided, the respective concave portions **24A** may be equidistantly arranged in the circumferential direction CD or may be arranged with different pitches. Also, the length of each concave portion **24A** in the circumferential direction CD is not limited to the length depicted in FIG. **10**. Also, the length of each concave portion **24A** in the circumferential direction CD may be equal as depicted in FIG. **10** or different. Also, in Preferred Embodiment 6, the depth (length in the Z-axis direction) of each concave portion **24A** is, for example, about 10 to about 100 (μm), but is not limited to this.

(127) The concave portion **24A** (in detail, a space defined by the concave portions **24A**) is one example of the communicating portion. A space defined by the concave portion **24A** is one example of a third space. It can also be said that paired inner side surfaces **24d** and **24e** and an inner bottom surface **24f** of the protruding portion **24**, which configure the concave portion **24A**, are one example of the communicating portion.

(128) According to Preferred Embodiment 6, by only providing the concave portions **24A** in the protruding portion **24**, the function of the communicating portion can be achieved.

Preferred Embodiment 7

(129) FIG. **12** is a perspective view of a lid member included in a semiconductor device according to Preferred Embodiment 7 of the present invention from a bottom surface side. FIG. **13** is a longitudinal sectional view of the semiconductor device according to Preferred Embodiment 7. A semiconductor device **10F** according to Preferred Embodiment 7 is different from the semiconductor device **10** according to Preferred Embodiment 1 in that the lid member **26** does not include the slit **27** but includes concave portions **26A**.

(130) As depicted in FIG. **12**, the lid member **26** of the semiconductor device **10F** does not include the slit **27** as depicted in FIG. **2**.

(131) The lid member **26** includes two concave portions **26A**. The concave portions **26A** each are one example of a second concave portion. Each concave portion **26A** is formed with the lower surface **26d** of the lid member **26** recessed upward.

(132) As depicted in FIG. **13**, each concave portion **26A** extends from the inner side surface **24c** of the protruding portion **24** of the resin package **22** to an exposure hole **22a** side in the Y-axis direction. With this, each concave portion **26A** communicates with the exposure hole **22a**. Each concave portion **26A** extends outward from the outer side surface **24a** of the protruding portion **24** in the Y-axis direction. With this, each concave portion **26A** communicates with the outside of the semiconductor device **10F**. As described above, via each concave portion **26A**, the exposure hole **22a** and the outside of the semiconductor device **10F** communicate with each other.

(133) In Preferred Embodiment 7, as depicted in FIG. **13**, the two concave portions **26A** are positioned on one straight line in the Y-axis direction. In other words, as depicted in FIG. **12**, the two concave portions **26A** are arranged with 180-degree pitches in the circumferential direction CD.

(134) Note that the number of concave portions **26A** included in the protruding portion **26** is not limited to two. Also, the position where each concave portion **26A** is located is not limited to the position depicted in FIG. **12**. Also, the two concave portions **26A** may be arranged in the circumferential direction CD with pitches other than 180-degree pitches. For example, between the two concave portions **26A**, a narrower pitch in the circumferential direction CD may be 90 degrees and a wider pitch in the circumferential direction CD may be 270 degrees. Also, when three or more concave portions **26A** are formed, the respective concave portions **26A** may be equidistantly arranged in the circumferential direction CD or may be arranged with different pitches. Also, the size and shape of each concave portion **26A** are not limited to the size and shape depicted in FIG. **12**. For example, in Preferred Embodiment 7, the depth of each concave portion **26A** is, about 10 to about 100 (μm), but is not limited to this. Also, the size and shape of each concave portion **26A** may be equal or different.

(135) The concave portion **26A** (in detail, a space defined by the concave portions **26A**) is one example of the communicating portion. A space defined by the concave portion **26A** is one example of a fourth space. It can also be said that an inner side surface **26e** and an inner bottom surface **26f**, of the lid member **26**, configuring the concave portion **26A** are one example of the communicating portion.

(136) According to Preferred Embodiment 7, by only providing the concave portions **26A** in the protruding portion **26**, the function of the communicating portion can be achieved.

Preferred Embodiment 8

(137) FIG. **14** is a plan view of a semiconductor device according to Preferred Embodiment 8 of the present invention. FIG. **15** is a sectional view along a D-D line of FIG. **14**. A semiconductor device **10G** according to Preferred Embodiment 8 is different from the semiconductor device **10** according to Preferred Embodiment 1 in that the lid member **26** includes the slit **27** and the protruding portion **24** of the resin package **22** includes the concave portions **24A**.

(138) As depicted in FIG. **14** and FIG. **15**, the lid member **26** of the semiconductor device **10G** includes two slits **27**. Note that the number, formation positions, shape, and size of the slits **27** included in the lid member **26** can be any.

(139) As with Preferred Embodiment 6, the protruding portion **24** of the resin package **22** of the semiconductor device **10G** includes two concave portions **24A**. Note that the number, formation positions, shape, and size of the concave portions **24A** included in the protruding portion **24** can be any.

(140) A communicating portion includes each slit **27** and each concave portion **24A**. In detail, the communicating portion includes a first space defined by each slit **27** and a third space defined by each concave portion **24A**.

(141) As depicted in FIG. **14**, in plan view, the slits **27** and the concave portions **24A** overlap each

other. That is, in plan view, the first space and the third space overlap each other. With this, as depicted in FIG. 15, the first space defined by each slit 27 and the third space defined by each concave portion 24A communicate with each other.

(142) In the semiconductor device 10G depicted in FIG. 14 and FIG. 15, the lid member 26 includes the slits 27, and the protruding portion 24 of the resin package 22 includes the concave portions 24A.

(143) However, the semiconductor device 10G according to Preferred Embodiment 8 may include, in addition to the slits 27 and the concave portions 24A or in place of the slits 27 and the concave portions 24A, at least one of the adhering portion 30 and the concave portions 26A. That is, the semiconductor device 10G according to Preferred Embodiment 8 is only required to include at least two of the slits 27, the adhering portion 30, the concave portions 24A, and the concave portions 26A. As described in Preferred Embodiment 5, the adhering portion 30 is interposed between the protruding portion 24 and the lid member 26. As described in Preferred Embodiment 7, the concave portions 26A are provided in the lower surface 26d of the lid member 26.

(144) When the lid member 26 of the semiconductor device 10G according to Preferred Embodiment 8 includes the slits 27, a communicating portion of the semiconductor device 10G includes a first space defined by each slit 27.

(145) When the semiconductor device 10G according to Preferred Embodiment 8 includes the adhering portion 30, the adhering portion 30 is not applied over the entire circumference of the upper surface 24b of the protruding portion 24 in plan view. In this case, the communicating portion of the semiconductor device 10G includes the space 31 (second space) provided in a portion to which the adhering portion 30 is not applied.

(146) When the protruding portion 24 of the resin package 22 of the semiconductor device 10G according to Preferred Embodiment 8 includes the concave portions 24A, the communicating portion of the semiconductor device 10G includes a third space defined by each concave portion 24A.

(147) When the lid member 26 of the semiconductor device 10G according to Preferred Embodiment 8 includes the concave portions 26A, the communicating portion of the semiconductor device 10G includes a fourth space defined by each concave portion 26A.

(148) As described above, the communicating portion of the semiconductor device 10G according to Preferred Embodiment 8 includes at least two spaces among the first space, the second space, the third space, and the fourth space. Said at least two spaces included in the communicating portion overlap each other in plan view. The first space to the fourth space may partially overlap each other or may entirely overlap each other.

(149) Note that all of the at least two spaces included in the communicating portion are not required to overlap each other in plan view. For example, when the communicating portion includes the first space to the third space, in plan view, while the first space and the second space overlap each other, the third space does not have to overlap the first second and the second space. Also, for example, when the communicating portion includes the first space to the fourth space, in plan view, while the first space and the second space overlap each other and the third space and the fourth space overlap each other, the first space and the second space do not have to overlap the third space and the fourth space.

(150) According to Preferred Embodiment 8, since two or more spaces among the first space to the fourth space overlap each other and communicate with each other in plan view, the cross-sectional area of the communicating portion can be increased. Since this facilitates circulation of the fluid in the communicating portion, responsiveness of the detecting unit 16d which detects pressure of the fluid can be improved.

(151) In the structure depicted in FIG. 14 and FIG. 15 described as one example of Preferred Embodiment 8, the first space and the third space each singly cause the exposure hole 22a and the outside of the semiconductor device 10G to communicate with each other. However, two or more

spaces among the first space to the fourth space may cooperate to cause the exposure hole **22a** and the outside of the semiconductor device **10G** to communicate with each other.

(152) FIG. **16** is a longitudinal sectional view of a modification of the semiconductor device according to Preferred Embodiment 8 of the present invention. For example, in a semiconductor device **10H** depicted in FIG. **16**, the slit **27** of the lid member **26** communicates with the outside of the semiconductor device **10H** but does not communicate with the exposure hole **22a**, and the concave portion **24A** of the protruding portion **24** of the resin package **22** communicates with the exposure hole **22a** but is not open sideward of the semiconductor device **10H**. However, in plan view, the slit **27** and the concave portion **24A** overlap each other, and thereby the slit **27** and the concave portion **24A** communicate with each other. With this, the exposure hole **22a** and the outside of the semiconductor device **10H** communicate with each other via the concave portion **24A** and the slit **27**.

Preferred Embodiment 9

(153) FIG. **17** is a plan view of a semiconductor device according to Preferred Embodiment 9 of the present invention. A semiconductor device **10I** according to Preferred Embodiment 9 is different from the semiconductor device **10G** according to Preferred Embodiment 8 in that the first space and the third space do not overlap each other in plan view.

(154) As depicted in FIG. **17**, the lid member **26** of the semiconductor device **10I** includes two slits **27**, and the protruding portion **24** of the resin package **22** of the semiconductor device **10I** includes two concave portions **24A**. In this point, the semiconductor device **10I** has the same structure as that of the semiconductor device **10G** according to Preferred Embodiment 8. However, in the semiconductor device **10I**, the two slits **27** and the two concave portions **24A** do not overlap each other in plan view. In this point, the semiconductor device **10I** is different from the semiconductor device **10G** according to Preferred Embodiment 8.

(155) As with the semiconductor device **10G** according to Preferred Embodiment 8, the semiconductor device **10I** according to Preferred Embodiment 9 is only required to include at least two of the slits **27**, the adhering portion **30**, the concave portions **24A**, and the concave portions **26A**. That is, a communicating portion of the semiconductor device **10I** according to Preferred Embodiment 9 is only required to include at least two spaces among the first space, the second space, the third space, and the fourth space. In this case, in Preferred Embodiment 9, the at least two spaces included in the communicating portion do not overlap each other in plan view.

(156) According to Preferred Embodiment 9, the number of paths through which the fluid circulates can be increased. Since this facilitates circulation of the fluid in the communicating portion, responsiveness of the detecting unit **16d** which detects pressure of the fluid can be improved.

(157) Note that by combining any preferred embodiments among the above-described various preferred embodiments as appropriate, the effects of each preferred embodiment can be achieved. For example, the lid member **26** of the semiconductor device **10B** according to Preferred Embodiment 3 may include a plurality of slits **28** (slits **28** each including the wide portion **28A** and the narrow portion **28B**) of Preferred Embodiment 2, or may include at least one of the slits **27** of Preferred Embodiment 1 and at least one of the slits **28** of Preferred Embodiment 2.

(158) While preferred embodiments of the present invention have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the present invention. The scope of the present invention, therefore, is to be determined solely by the following claims.

Claims

1. A semiconductor device which detects pressure, the semiconductor device comprising: a base substrate; a detector mounted on an upper surface of the base substrate and including a pressure

detector to detect pressure; a resin package provided on the upper surface of the base substrate, with the detector buried therein, and including a cavity which exposes the pressure detector upward; a lid supported by the resin package so as to close the cavity; and a communicating portion to cause the cavity to communicate with outside of the semiconductor device; wherein the resin package includes: a base portion where the detector is buried; and a protruding portion protruding upward from the base portion and including the cavity; in plan view, the base portion extends outward from an outer side surface of the protruding portion; the lid is supported by an upper surface of the protruding portion; the lid includes an extending portion in at least a portion of an outer circumferential portion of the lid, the extending portion extending outward from the outer side surface of the protruding portion in plan view; and the communicating portion is open sideward of the semiconductor device to communicate with outside.

2. The semiconductor device according to claim 1, wherein the communicating portion includes a space defined by a slit located inward of the outer circumferential portion of the lid in plan view.

3. The semiconductor device according to claim 2, wherein the slit is at a position outside of the pressure detector in plan view.

4. The semiconductor device according to claim 2, wherein the slit includes: a wide portion at least a portion of which overlaps the cavity in plan view; and a narrow portion connected to the wide portion and not overlapping the cavity in plan view.

5. The semiconductor device according to claim 2, wherein the lid includes three or more of the slits.

6. The semiconductor device according to claim 1, further comprising: an adhering portion interposed partially between the upper surface of the protruding portion and the lid to fix the lid to the protruding portion; wherein the communicating portion includes a space between the upper surface of the protruding portion and the lid which are opposed to each other in a portion between the upper surface of the protruding portion and the lid where the adhering portion is not interposed.

7. The semiconductor device according to claim 1, wherein the protruding portion includes a first concave portion recessed downward from the upper surface of the protruding portion; and the communicating portion includes a space defined by the first concave portion.

8. The semiconductor device according to claim 1, wherein the lid includes a second concave portion recessed upward from a lower surface of the lid; and the communicating portion includes a space defined by the second concave portion.

9. The semiconductor device according to claim 1, further comprising: at least two of a slit located inward of the outer circumferential portion of the lid in plan view, an adhering portion interposed partially between the upper surface of the protruding portion and the lid to fix the lid to the protruding portion, a first concave portion recessed downward from the upper surface of the protruding portion, and a second concave portion recessed upward from a lower surface of the lid; wherein when the semiconductor device includes the slit, the communicating portion includes a first space defined by the slit; when the semiconductor device includes the adhering portion, the communicating portion includes a second space between the upper surface of the protruding portion and the lid which are opposed to each other in a portion between the upper surface of the protruding portion and the lid where the adhering portion is not provided; when the semiconductor device includes the first concave portion, the communicating portion includes a third space defined by the first concave portion; when the semiconductor device includes the second concave portion, the communicating portion includes a fourth space defined by the second concave portion; and in plan view, the first space, the second space, the third space, and the fourth space overlap one another.

10. The semiconductor device according to claim 1, further comprising: at least two of a slit located inward of the outer circumferential portion of the lid in plan view, an adhering portion interposed partially between the upper surface of the protruding portion and the lid to fix the lid to the protruding portion, a first concave portion recessed downward from the upper surface of the

protruding portion, and a second concave portion recessed upward from a lower surface of the lid; wherein when the semiconductor device includes the slit, the communicating portion includes a first space defined by the slit; when the semiconductor device includes the adhering portion, the communicating portion includes a second space between the upper surface of the protruding portion and the lid which are opposed to each other in a portion between the upper surface of the protruding portion and the lid where the adhering portion is not provided; when the semiconductor device includes the first concave portion, the communicating portion includes a third space defined by the first concave portion; when the semiconductor device includes the second concave portion, the communicating portion includes a fourth space defined by the second concave portion; and in plan view, the first space, the second space, the third space, and the fourth space do not overlap one another.

11. The semiconductor device according to claim 1, further comprising: a circuit element mounted on the upper surface of the base substrate; and a connector electrically connecting the circuit element and the detector together.

12. An electronic apparatus comprising: the semiconductor device according to claim 1; an O ring surrounding the protruding portion of the resin package of the semiconductor device in plan view and including an inner circumferential portion in contact with the outer side surface of the protruding portion; and a casing to which the semiconductor device is attached.

13. The electronic apparatus according to claim 12, wherein the communicating portion includes a space defined by a slit located inward of the outer circumferential portion of the lid in plan view.

14. The electronic apparatus according to claim 13, wherein the slit is at a position outside of the pressure detector in plan view.

15. The electronic apparatus according to claim 13, wherein the slit includes: a wide portion at least a portion of which overlaps the cavity in plan view; and a narrow portion connected to the wide portion and not overlapping the cavity in plan view.

16. The electronic apparatus according to claim 13, wherein the lid includes three or more of the slits.

17. The electronic apparatus according to claim 12, further comprising: an adhering portion interposed partially between the upper surface of the protruding portion and the lid to fix the lid to the protruding portion; wherein the communicating portion includes a space between the upper surface of the protruding portion and the lid which are opposed to each other in a portion between the upper surface of the protruding portion and the lid where the adhering portion is not interposed.

18. The electronic apparatus according to claim 12, wherein the protruding portion includes a first concave portion recessed downward from the upper surface of the protruding portion; and the communicating portion includes a space defined by the first concave portion.

19. The electronic apparatus according to claim 12, wherein the lid includes a second concave portion recessed upward from a lower surface of the lid; and the communicating portion includes a space defined by the second concave portion.

20. The electronic apparatus according to claim 12, further comprising: a circuit element mounted on the upper surface of the base substrate; and a connector electrically connecting the circuit element and the detector together.
